رواد مصر الرقمية

وزارة الاتصالات
وتكنولوجيا المعلومات

Heart Disease Risk Prediction

DEPI Final Project - Round 3

September 2025

Contents

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2.Project Information

Dataset	**https://www.kaggle.com/datasets/alexteboul/heart-disease-health-indicators-dataset**
GitHub Repo	https://github.com/ahmed2005hussen/DEPI_Final_Project
Application	https://depi-final-project.streamlit.app/
Dashboard	https://github.com/ahmed2005hussen/DEPI_Final_Project/blob/main/Heart%20Disease%20Prediction%20Dashboard.pdf
Notebook	https://github.com/ahmed2005hussen/DEPI_Final_Project/blob/main/main.ipynb

3. Project Statement

Heart disease is one of the most common diseases in the world and often leads to **death**. Many people **do not know** they have heart disease until they experience a **heart attack** or **stroke**. Most people are also **unwilling** to visit doctors regularly to monitor their health.

However, with the advancement of technology, we can help these people by providing them with a **prediction** about their health condition **without** the need to **visit** a doctor. From **home**, at **any time**, they can simply open their **phone**, answer a few questions, and instantly check their health status.

Since people are naturally worried and always have questions, we also need to **integrate** a **chatbot** that can answer their concerns almost like having a **private doctor** available at any moment.

Our goal is to simplify this process and enable people to check their health status without frequent doctor visits, with no extra cost, no effort, and most importantly, to help them live a healthier life.

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4. Motivation & Objectives

4.1 Why these matters

This project will help people take better care of their health.

Benefits include:

- **Save lives**: Early detection can prevent heart attacks.
- **Save money**: Prevention is cheaper than treatment.
- **Help doctors**: Provide healthcare providers with a quick screening tool.
- **Easy to use**: Accessible to the public.
- **High value**: Can save people's lives.
- **Recommendation**: Chat boot can answer people's questions.

4.2 Project goals:

- Develop a **predictive analytics system** to assess patient health risk and provide clear, explainable insights into their medical status.
- Enable **personalized healthcare support** through recommendations, AI-powered health coaching and lifestyle guidance.
- Support proactive healthcare with the **automated health plans**, real-time risk classification and preventive recommendations.
- Promote general health awareness by delivering accessible advice for maintaining a **healthy lifestyle**.

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5. Project stages

5.1 Collect Data

This is the first and most important stage. We spent a lot of time searching for a suitable dataset, and eventually we found one that meets our needs. It contains many records, which makes it suitable for training our model and achieving reliable prediction results.

Dataset Source: Kaggle - Heart Disease Health Indicators Dataset.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
HeartDisea	HighBP	HighChol	CholCheck	BMI	Smoker	Stroke	Diabetes	PhysActivit	Fruits	Veggies	HvyAlcoho	AnyHealth	NoDocbcC	GenHlth	MentHlth	PhysHlth	DiffWalk	Sex	Age	Education	Income
0	1	1	1	40	1	0	0	0	0	1	0	1	0	5	18	15	1	0	9	4	3
0	0	0	0	25	1	0	0	1	0	0	0	0	1	3	0	0	0	0	7	6	1
0	1	1	1	28	0	0	0	0	1	0	0	1	1	5	30	30	1	0	9	4	8
0	1	0	1	27	0	0	0	1	1	1	0	1	0	2	0	0	0	0	11	3	6
0	1	1	1	24	0	0	0	1	1	1	0	1	0	2	3	0	0	0	11	5	4
0	1	1	1	25	1	0	0	1	1	1	0	1	0	2	0	2	0	1	10	6	8
0	1	0	1	30	1	0	0	0	0	0	0	1	0	3	0	14	0	0	9	6	7
0	1	1	1	25	1	0	0	1	0	1	0	1	0	3	0	0	1	0	11	4	4
1	1	1	1	30	1	0	2	0	1	1	0	1	0	5	30	30	1	0	9	5	1
0	0	0	1	24	0	0	0	0	0	1	0	1	0	2	0	0	0	1	8	4	3
0	0	0	1	25	1	0	2	1	1	1	0	1	0	3	0	0	0	1	13	6	8
0	1	1	1	34	1	0	0	0	1	1	0	1	0	3	0	30	1	0	10	5	1
0	0	0	1	26	1	0	0	0	0	1	0	1	0	3	0	15	0	0	7	5	7
0	1	1	1	28	0	0	2	0	0	1	0	1	0	4	0	0	1	0	11	4	6
0	0	1	1	33	1	1	0	1	0	1	0	1	1	4	30	28	0	0	4	6	2
0	1	0	1	33	0	0	0	1	0	0	0	1	0	2	5	0	0	0	6	6	8
0	1	1	1	21	0	0	0	1	1	1	0	1	0	3	0	0	0	0	10	4	3
0	0	0	1	23	1	0	2	1	0	0	0	1	0	2	0	0	0	1	7	5	6
0	0	0	0	23	0	0	0	0	0	1	0	1	0	2	15	0	0	0	2	6	7
0	0	1	1	28	0	0	0	0	0	0	1	1	0	2	10	0	0	1	4	6	8
1	1	1	1	22	0	1	0	0	1	0	0	1	0	3	30	0	1	0	12	4	4
0	1	1	1	38	1	0	0	0	1	1	0	1	0	5	15	30	1	0	13	2	3
0	0	0	1	28	1	0	0	0	0	1	0	1	0	3	0	7	0	1	5	5	5
0	1	0	1	27	0	0	2	1	1	1	0	1	0	1	0	0	0	0	13	5	4
0	1	1	1	28	1	0	0	0	1	1	0	1	0	3	6	0	1	0	9	4	6

5.2 Data Understanding:

Once we obtain the dataset, we will thoroughly analyze it to understand its structure and content. This includes identifying which features to keep or remove, interpreting the meaning of each value, assessing feature importance for the prediction task, and determining how to preprocess and engineer features to extract useful information.

Key properties: 253680 rows- 22 columns - No missing values - Mix of binary, categorical (age groups), and numeric (BMI) features.

5.2.1 Main Features Description

Column	Description
HeartDiseaseorAttack	Describes if person have a heart disease or not
HighBP	Describes if person have a high blood pressure or not
HighChol	Describes if person have a high cholesterol or not
BMI	Body Mass Index
Smoker	If person smoke or not
Stroke	If person had a stroke before or not
Diabetes	If person have diabetes or not
PhysActivity	If person play sports or not
GenHlth	Rate for general health
MentHlth	Number of days with bad mental health
PhysHlth	Number of days with bad physical health
DiffWalk	If person have a difficulty in walking
Sex	Person's sex (male or female)
Age	Not exact age, categorical variable coded from 1 to 13 (each = age group)

Code	Age Group
1	18–24
2	25–29
3	30–34
4	35–39
5	40–44
6	45–49
7	50–54
8	55–59
9	60–64
10	65–69
11	70–74
12	75–79
13	80+

5.2.2 Data information

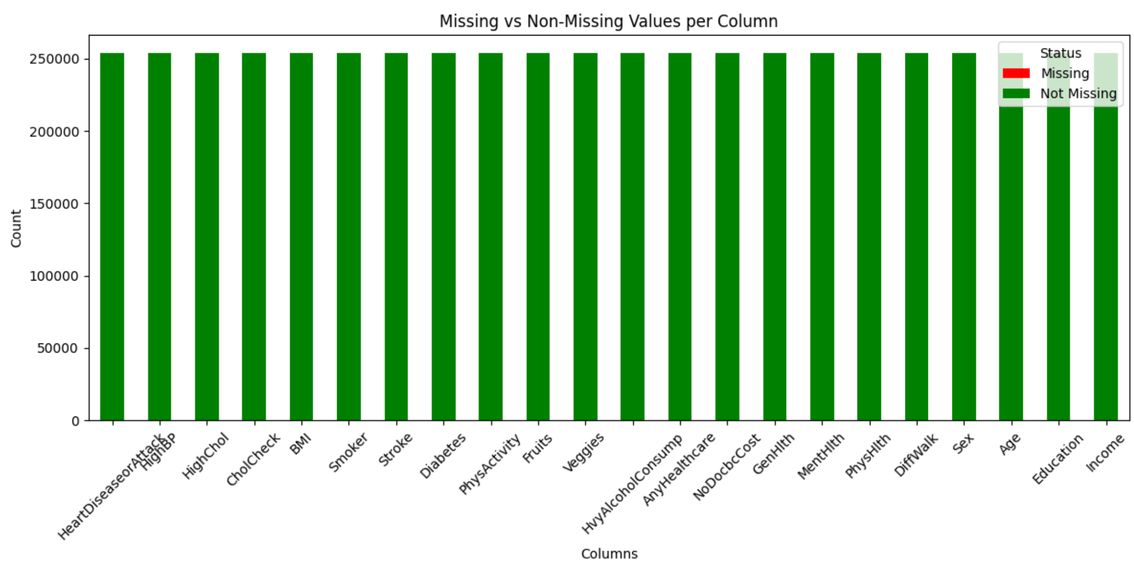
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 253680 entries, 0 to 253679
Data columns (total 22 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   HeartDiseaseorAttack  253680 non-null  float64
 1   HighBP                253680 non-null  float64
 2   HighChol              253680 non-null  float64
 3   CholCheck             253680 non-null  float64
 4   BMI                   253680 non-null  float64
 5   Smoker                253680 non-null  float64
 6   Stroke                253680 non-null  float64
 7   Diabetes              253680 non-null  float64
 8   PhysActivity          253680 non-null  float64
 9   Fruits                253680 non-null  float64
 10  Veggies               253680 non-null  float64
 11  HvyAlcoholConsump     253680 non-null  float64
 12  AnyHealthcare         253680 non-null  float64
 13  NoDocbcCost           253680 non-null  float64
 14  GenHlth               253680 non-null  float64
 15  MentHlth              253680 non-null  float64
 16  PhysHlth              253680 non-null  float64
 17  DiffWalk              253680 non-null  float64
 18  Sex                   253680 non-null  float64
 19  Age                   253680 non-null  float64
 20  Education             253680 non-null  float64
 21  Income                253680 non-null  float64
dtypes: float64(22)
memory usage: 42.6 MB
```

Check if have any Missing values

```
df.isnull().sum()
```

```
HeartDiseaseorAttack    0
HighBP                  0
HighChol                0
CholCheck               0
BMI                     0
Smoker                  0
Stroke                  0
Diabetes                0
PhysActivity            0
Fruits                  0
Veggies                 0
HvyAlcoholConsump       0
AnyHealthcare           0
NoDocbcCost             0
GenHlth                 0
MentHlth                0
PhysHlth                0
DiffWalk                0
Sex                     0
Age                     0
Education               0
Income                  0
dtype: int64
```



As you see our data is clean, we don’t have null value.

5.2.3 Environment & Dependencies

Primary libraries: - Python- pandas - numpy - matplotlib - seaborn - scikit-learn - imbalanced-learn (SMOTE) - xgboost – joblib – (deployment) streamlit – (chat boot) Groq

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5.2.4 Data Exploration & Preprocessing

Key steps performed:

1. Loaded CSV and printed `df.info()` and `df.describe()` to inspect types and distributions.

2. Verified there are no null values: `df.isnull().sum()` returned zeros for all columns.

3. Investigated duplicates kept duplicates because dataset lacks unique ID and duplicates may represent valid entries.

4. Studied BMI distribution; observed outliers in the upper tail and decided to retain them as medically relevant cases.

5. Converted several binary columns to boolean dtype (e.g., `HighBP`, `Smoker`, `Diabetes`, `Sex`, `HighChol`). 6. Removed several non-essential columns and exported the cleaned CSV as `Dataset/Train.csv`.

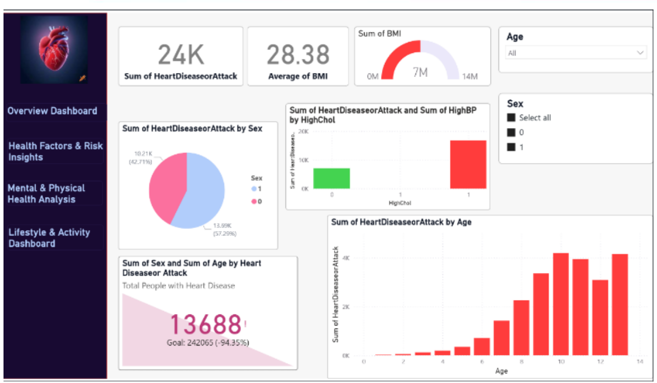
Imbalance handling: - The target is imbalanced; applied SMOTE to the training set after the train/test split to balance classes.

5.2.5 Feature Engineering

- Binned BMI into ranges for visualization and exploratory statistics.
- Left Age as an ordered categorical code (1..13) consistent with dataset documentation.
- Considered and removed low-importance features based on feature-importance analysis (e.g., `CholCheck`, `BMI`, `MentHlth`, `PhysHlth`) in an experimental run to reduce dimensionality.

5.3 Dashboard

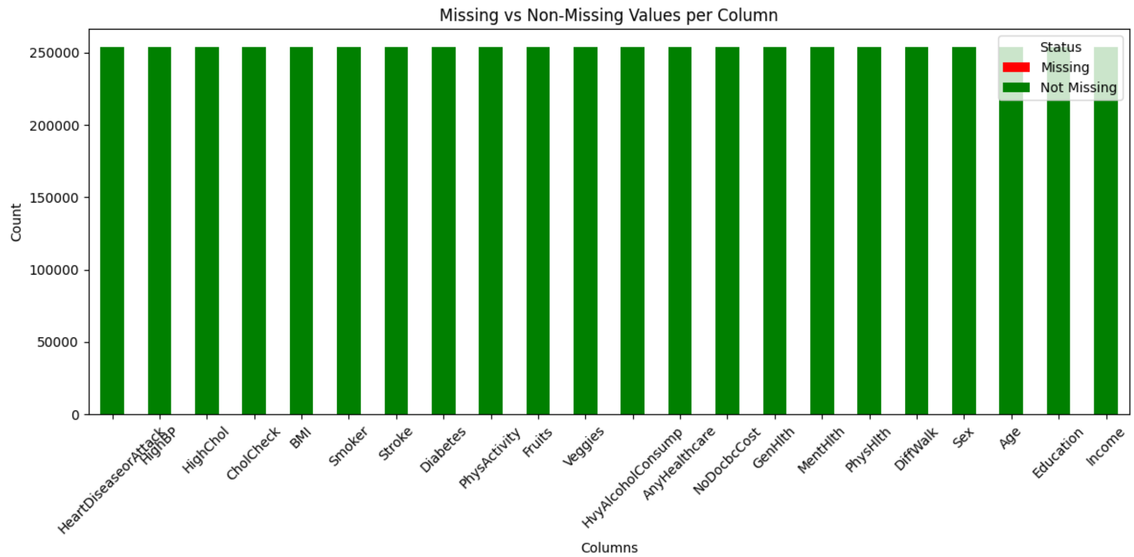
We have a beautiful dashboard to visualize our data.



https://github.com/ahmed2005hussen/DEPI_Final_Project/blob/main/Heart%20Disease%20Prediction%20Dashboard.pdf

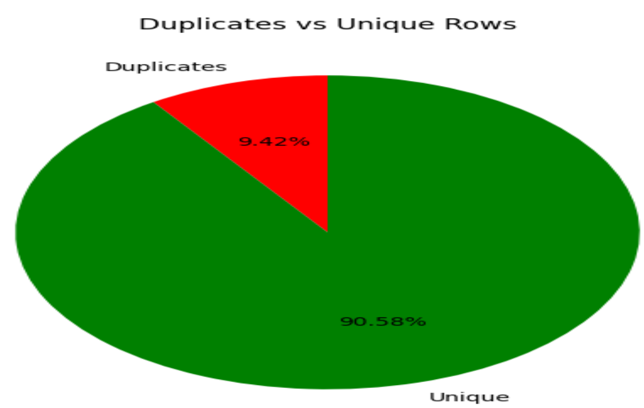
5.4 Exploratory Data Analysis (EDA) & Visualization

The chart shows that all columns in the dataset have zero missing values Each column has the same count of non-missing records (about 255,000), which means the dataset is complete and consistent with no need for imputation or handling of null values.



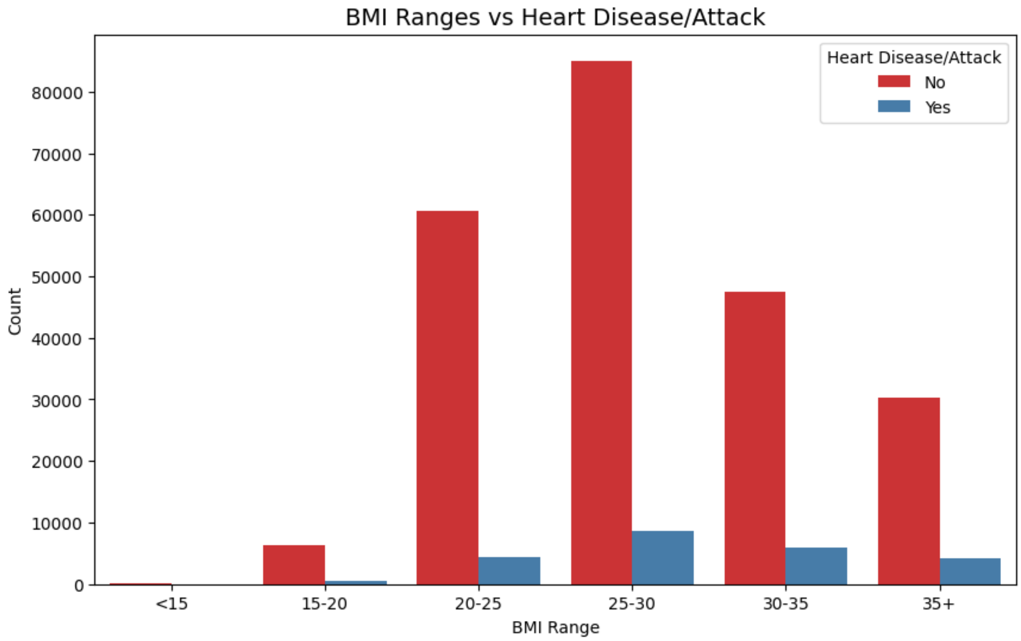
Duplicates and unique values

All columns have complete data with no missing values, ensuring dataset reliability for further analysis.



The dataset does not include a unique identifier to distinguish whether records belong to the same individual or different individuals. Therefore, duplicate rows may still represent valid cases and have been retained.

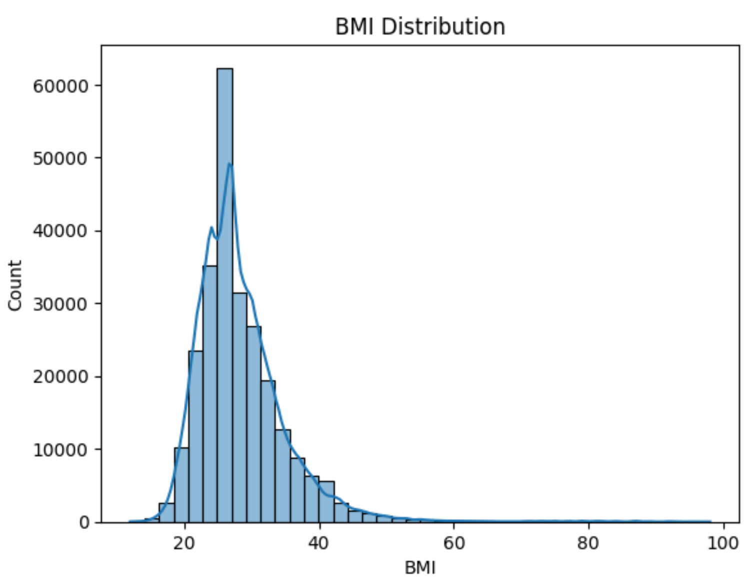
Relation Between BMI and Heart Disease



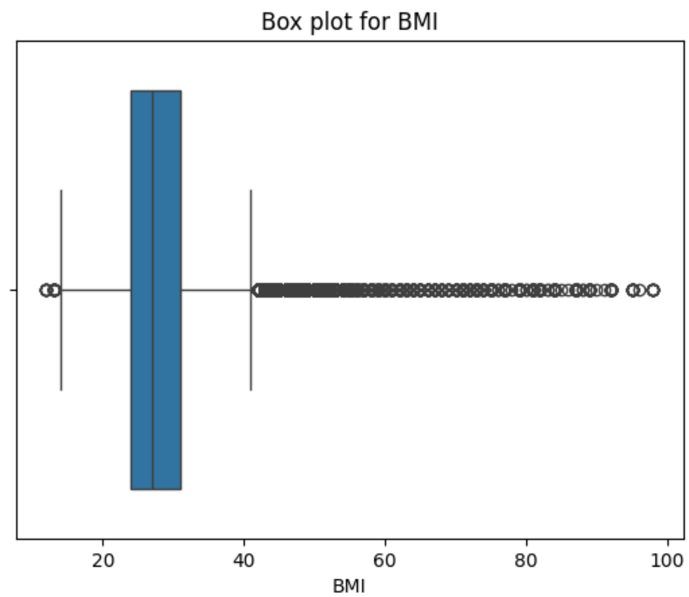
Most individuals fall within the BMI ranges of 20–30. However, the proportion of people with Heart Disease increases noticeably in higher BMI ranges (30+).

This suggests a positive relationship between obesity and the likelihood of heart disease or attack.

BMI Distribution

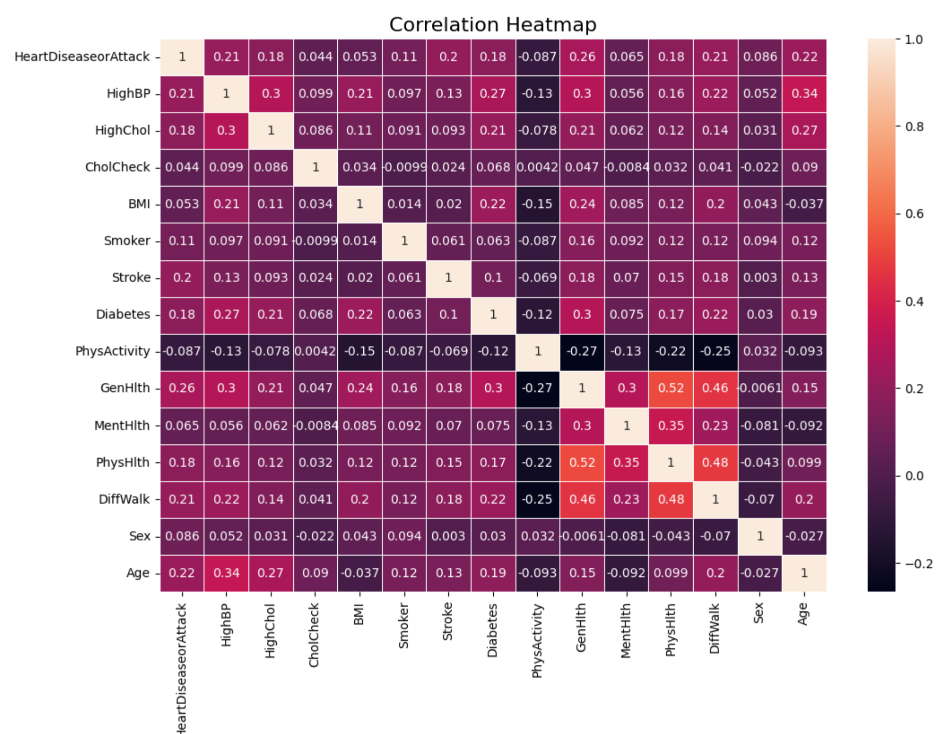


Box Plot for BMI

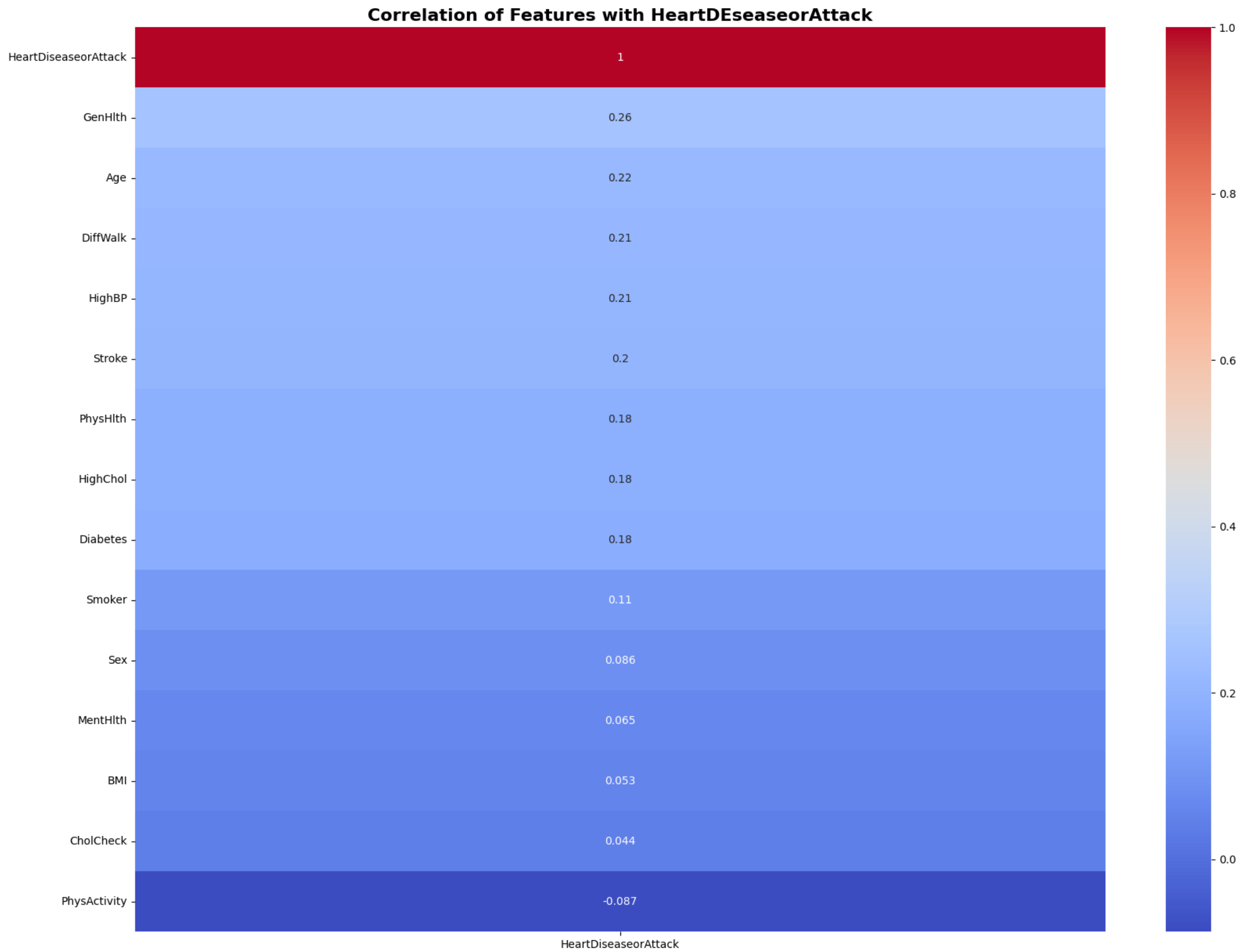


The box plot for BMI shows that while most individuals fall between Q1 ≈ 24 and Q3 ≈ 32, there are many outliers above the upper whisker (~40). These outliers represent individuals with significantly higher BMI values, and they are important to retain since they capture cases of severe obesity, which are strongly linked to health risks such as heart disease, diabetes, and high blood pressure.

Correlation



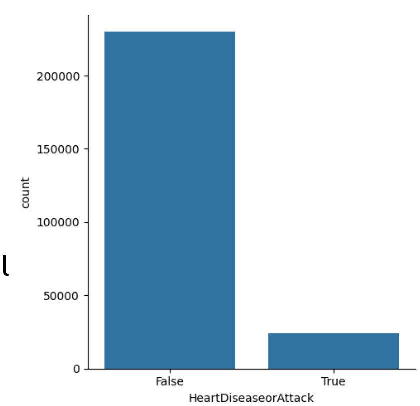
Correlation with the target:



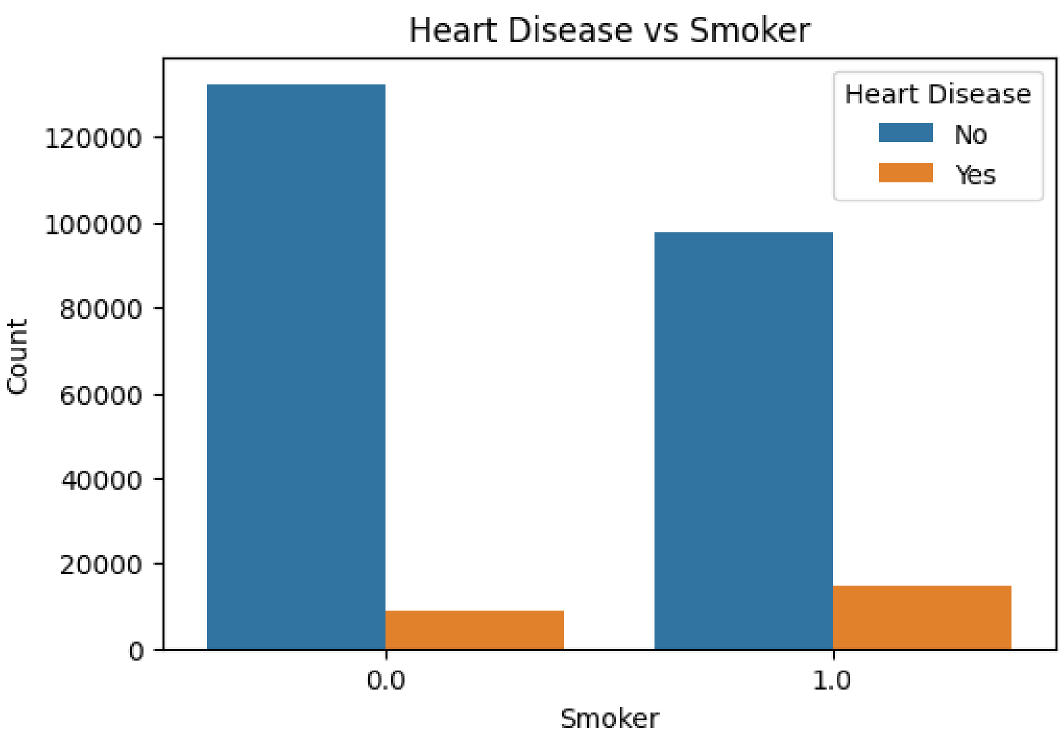
Count HeartDissesorAttack:

This indicates that our dataset is imbalanced,

which needs to be addressed before building the model



Heart Disease with Smokers:



Smoking appears to be a risk factor for heart disease in your dataset. While the absolute number of heart disease cases is smaller than non-cases, smokers have a higher relative proportion of heart disease than non-smokers.

5.5 Solve imbalance

Before modeling we need to solve the imbalance in our data, to avoid overfitting.

We used SMOTE to solve the imbalance:

```
from imblearn.over_sampling import SMOTE
from sklearn.model_selection import train_test_split
```

```
x = df1.drop("HeartDiseaseorAttack" , axis=1)

y = df1["HeartDiseaseorAttack"]
```

```
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
print("Before SMOTE:\n", y_train.value_counts())
```

```
Before SMOTE:
 HeartDiseaseorAttack
False    183819
True      19125
Name: count, dtype: int64
```

```
smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)
print("After SMOTE:\n", y_train_res.value_counts())
```

```
After SMOTE:
 HeartDiseaseorAttack
False    183819
True     183819
Name: count, dtype: int64
```

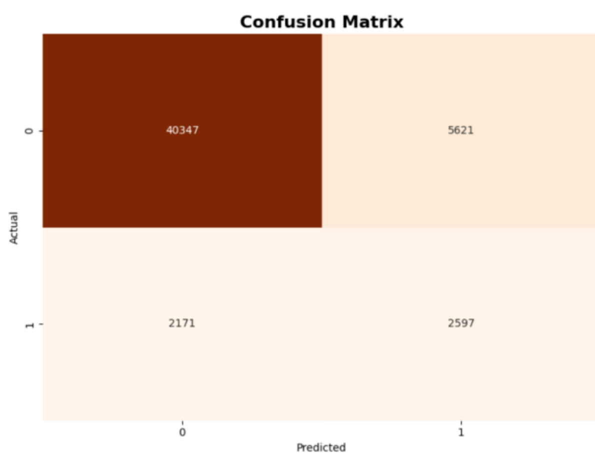
5.6 Modeling

We chose the **XGBoost** classifier for our project because it is known for its high accuracy, efficiency, and strong performance in classification tasks. XGBoost handles large datasets effectively and captures complex patterns in the data, which is essential for predicting heart disease. It also includes regularization techniques that reduce overfitting, making the model more reliable. Additionally, XGBoost can deal with missing values automatically and provides useful insights into feature importance, helping us understand which factors have the greatest impact on the prediction.

Before reaching the final model, we passed through three major improvement stages. Each stage focused on enhancing a specific aspect of the model, such as data quality, feature selection, and parameter optimization. The following section explains each stage in detail.

5.6.1 First Stage:

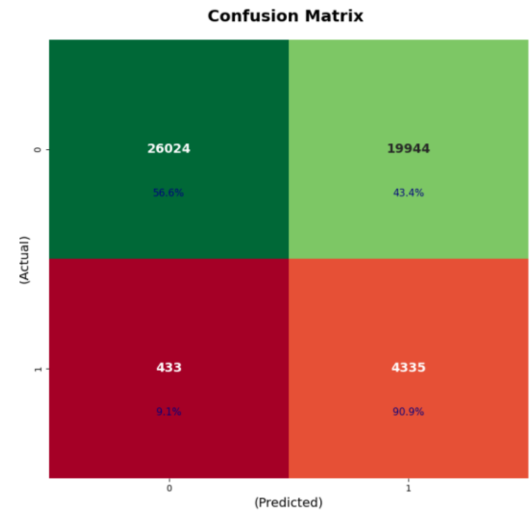
In the first stage, we trained a basic **XGBoost** classifier using default parameters to establish an initial performance baseline. The model achieved an accuracy of **84.64%**, with strong performance on healthy cases but weaker detection of heart disease cases.



The confusion matrix shows that the model correctly identified **2,597** patients but missed **2,171** (false negatives). It also produced **5,621 false positives**, indicating a trade-off between sensitivity and specificity. Overall, this stage highlighted the need for further improvement to enhance the model's ability to detect true heart disease cases more accurately. And that's meaning the model say “2171 is healthy“, but the truth they not. For this we need to reduce this number

5.6.2 Second Stage:

In the second stage, we focused on improving the model's recall for heart disease cases. Several hyperparameters were adjusted, including **increasing** the number of **trees**, reducing the learning rate, increasing model depth, adding subsampling, and applying class weighting with **scale_pos_weight=10** to emphasize the minority class.

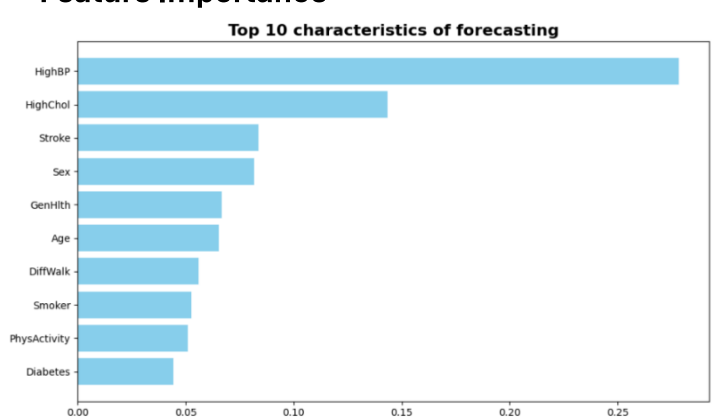


The confusion matrix shows that the model correctly identified 4,335 out of 4,768 patients while missing only 433 (a major improvement in patient safety).

However, this improvement came with a trade-off: accuracy dropped to 59.84%, and the number of false positives increased sharply (19,944 healthy individuals incorrectly flagged as at-risk). Despite this decrease in overall accuracy, the model successfully achieved the primary medical objective of minimizing missed heart disease cases.

Our goal is to discover the patient, the model now say to 433 patients you are healthy but they are not, but if you see the number now is less than the first stage, and this is good.

Feature Importance

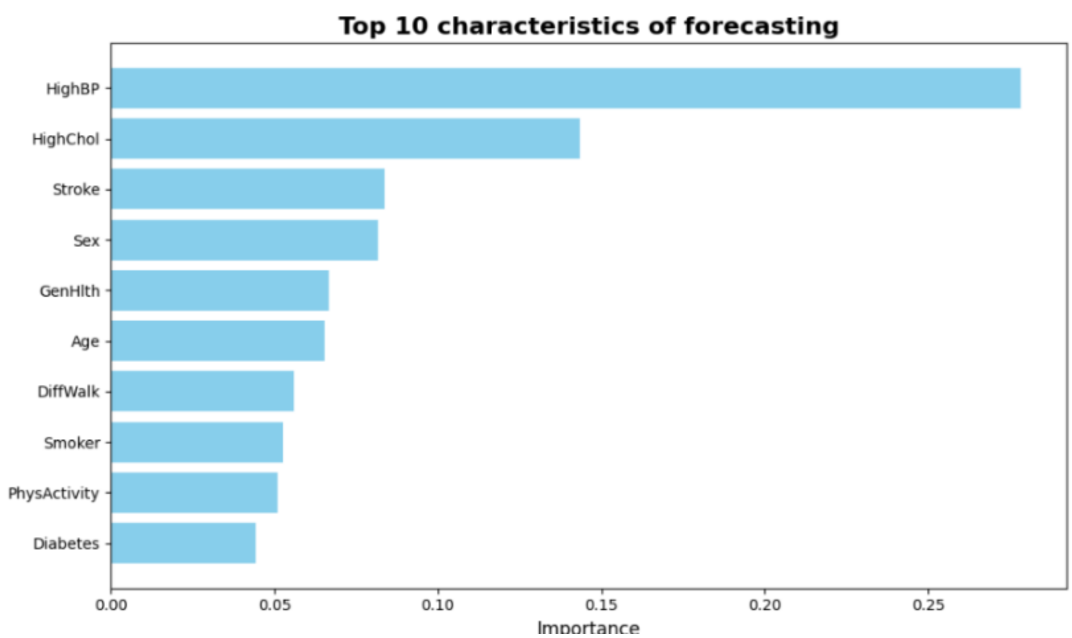


This is the top characteristics of forecasting so we can use it only for our model and see if the result is good.

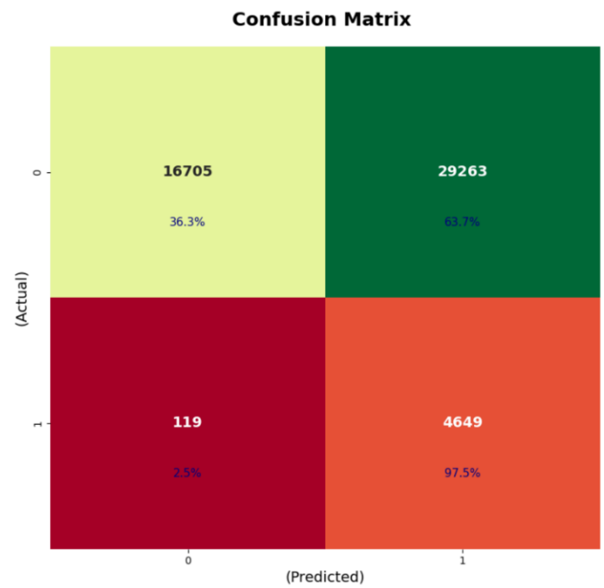
5.6.3 Third stage

In the third stage, we focused on strengthening the model's ability to detect heart disease cases by giving more attention to the minority class. To achieve this, several improvements were introduced before retraining the model. These included increasing the number of estimators, reducing the learning rate for more stable learning, increasing model depth, applying subsampling, and using class weighting with scale_pos_weight=10. Additionally, we used **SMOTE** to oversample the minority class, giving the model a balanced and representative training set.

After applying these enhancements, we **retrained the model** on the updated dataset and evaluated its performance on the test set.



Performance Evaluation



The confusion matrix shows that the model successfully identified **4,649** out of **4,768** true patients, missing only **119** cases. This represents a strong improvement in patient safety, as the number of missed cases is now significantly lower compared to previous stages.

However, this improvement came with a trade-off: the number of false positives increased (29,263 healthy individuals incorrectly classified as at-risk), leading to a drop in overall accuracy to 42.09%.
Even with this drop, the model achieved its most critical medical objective minimizing false negatives and ensuring that nearly all positive cases are detected.

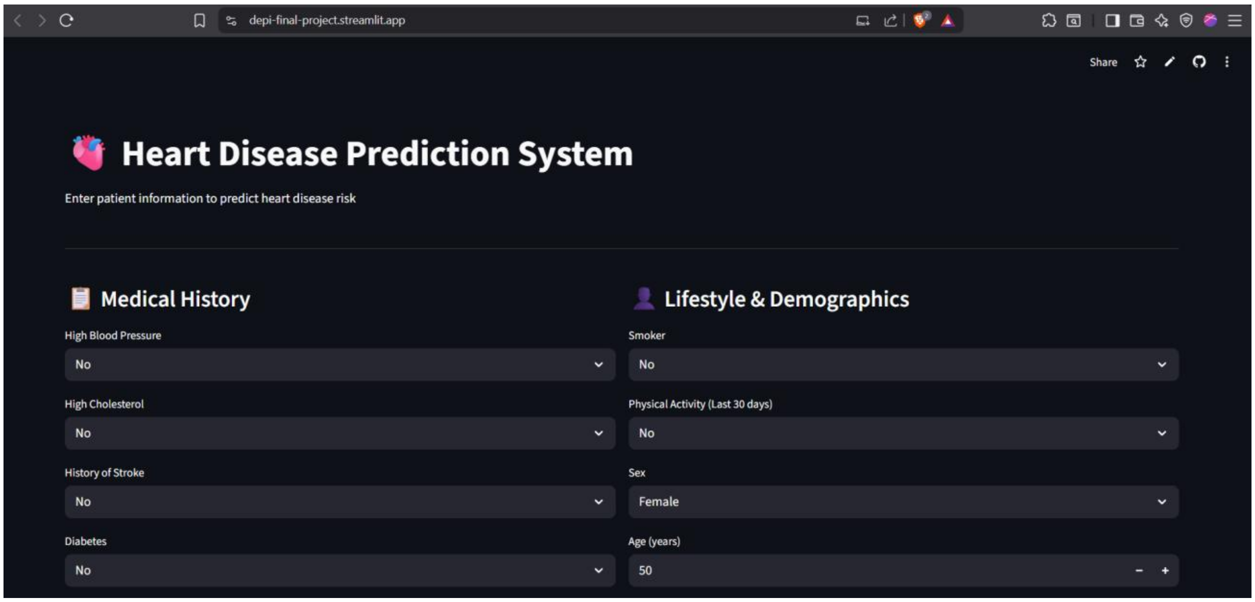
Our priority is to discover every potential patient. The model now misclassifies only 119 true cases as healthy, which is much lower than in earlier stages, and this is a positive and important step toward building a reliable early-warning system.

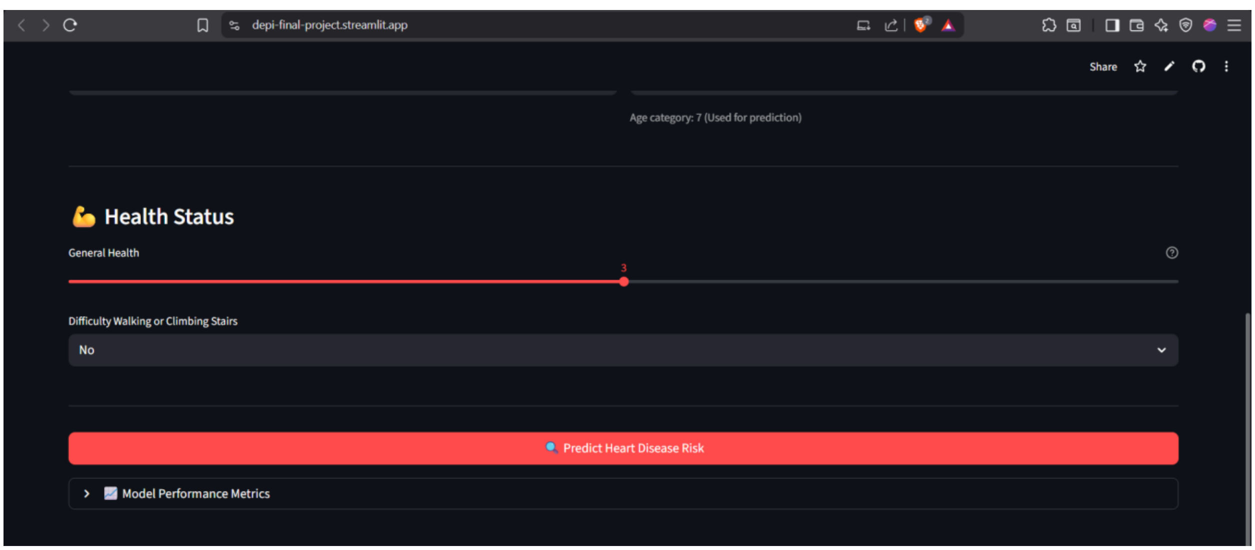
And this is our last model.

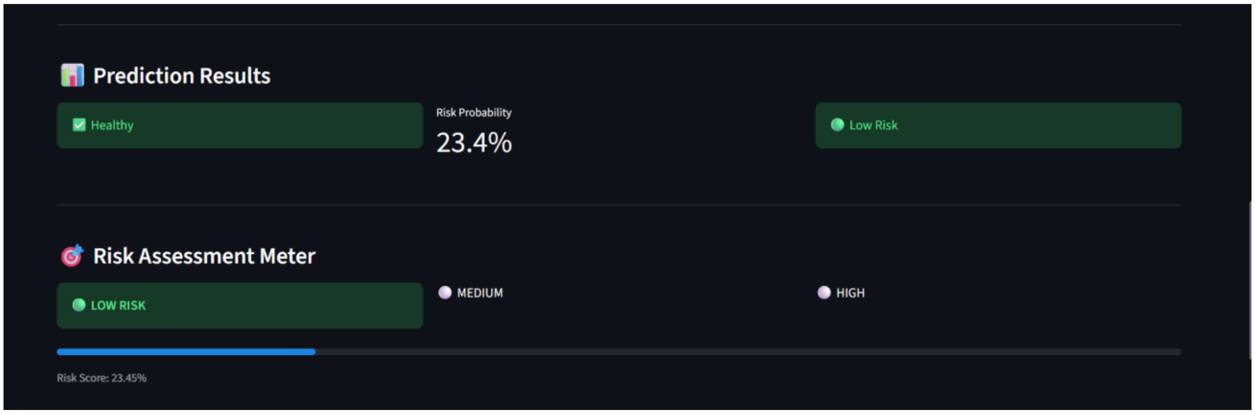
5.7 Deployment

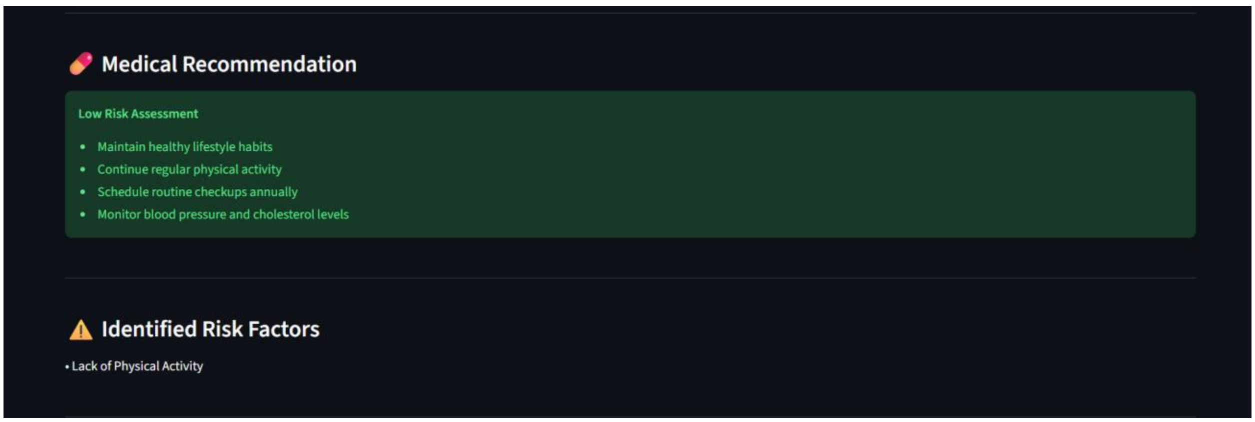
We use streamlit to deploy our model.

Link: https://depi-final-project.streamlit.app/






Prediction Results
Healthy
Risk Probability
23.4%
Low Risk
Risk Assessment Meter
LOW RISK
MEDIUM
HIGH
Risk Score: 23.45%


Medical Recommendation
Low Risk Assessment
Maintain healthy lifestyle habits
Continue regular physical activity
Schedule routine checkups annually
Monitor blood pressure and cholesterol levels
Identified Risk Factors
• Lack of Physical Activity

Chat boot


هل في خطر عليا؟
لا، لا توجد خطورة عالية.
نسبة خطرك الآن 23.4%، وهي ضمن الفئة «منخفضة»، ولا يُظهر التحليل وجود أمراض قلبية حاليًا.
لكن لأنك لا تمارسين نشاطًا بدنيًا حاليًا، فإن إضافة نصف ساعة من المشيط السريع أو أي نشاط تعرفينه (5 أيام في الأسبوع) يُقلّل الخطر إلى النصف تقريبًا، ويُحسّن صحة القلب والأوعية الدموية بشكل عام. لا يُوجد سبب آخر يستدعي القلق في هذه اللحظة، لكن يُفضّل إجراء فحص دوري للضغط والسكر والكوليسترول مرة سنويًا، ومراجعة طبيبك الخاص لو لاحظتِ أي أعراض جديدة.
تنويه: هذه المعلومات للتوعية فقط ولا تُعدُّ بديلاً عن استشارة طبيبك المباشرة.
Clear Chat History
Model Performance Metrics
Ask a question about your results...